CHAPTER ONE

1.0 Introduction and Opportunity Statement

The rapid evolution of information and communication technologies (ICT) has significantly transformed the way financial transactions are conducted across the globe. Over the past decade, mobile technologies have emerged as powerful tools for delivering financial services efficiently and securely. Mobile payment systems, in particular, have gained widespread acceptance due to their ability to facilitate instant transactions, reduce operational costs, and improve accessibility to financial services (Dahlberg, Guo, & Ondrus, 2015). The proliferation of smartphones, improved internet connectivity, and advancements in financial technology (FinTech) have accelerated the adoption of digital payment platforms in both developed and developing economies.

In developing countries such as Ghana, mobile money services have played a critical role in promoting financial inclusion. Mobile financial platforms have enabled individuals to send, receive, and store money without the need for traditional banking infrastructure (Aker & Mbiti, 2010). According to the Bank of Ghana (2023), mobile money transactions continue to grow significantly each year, demonstrating increased public trust and reliance on digital financial services. This digital transformation presents enormous opportunities for institutions, including universities, to modernize their financial operations.

Despite the increasing adoption of digital payment systems in commerce and personal finance, many higher educational institutions continue to rely on traditional and semi-manual methods for fee collection. These methods often require students to physically visit banks, stand in long queues, and submit printed payment receipts for verification. Such processes are not only time-consuming but also inefficient and vulnerable to human error. During peak academic periods such as registration and examination sessions, congestion at banks and administrative offices becomes a recurring challenge.

At Koforidua Technical University (KTU), students are required to complete fee payments through designated financial institutions before academic registration can be finalized. The current process involves multiple steps, including bank deposits, receipt collection, and manual verification by the university's finance department. This approach creates administrative bottlenecks, increases workload for staff, and may lead to delays in academic activities. Furthermore, students face risks associated with carrying large sums of cash and the possibility of losing or damaging physical receipts.

Research indicates that ease of use, perceived security, and system reliability are critical factors influencing user adoption of mobile payment systems (Oliveira et al., 2016; Zhou, 2013). When properly implemented, mobile payment applications enhance transaction transparency, improve financial record accuracy, and provide real-time confirmation of payments. These advantages make mobile payment integration highly suitable for educational institutions seeking to improve service delivery and operational efficiency.

The growing penetration of smartphones among university students presents a strategic opportunity to introduce a centralized mobile-based payment system tailored specifically for institutional needs. By leveraging existing mobile money infrastructure and secure payment gateways, Koforidua Technical University can transition from a predominantly manual payment system to a more automated and integrated digital solution. Such innovation aligns with global trends in digital transformation and e-governance within higher education.

This research therefore proposes the development of a Student Pay Mobile Application designed to streamline fee payments, generate instant digital receipts, and maintain secure transaction records for students of KTU. The proposed system will not only address current inefficiencies but also provide a scalable framework for future integration with other campus services. Ultimately, the project seeks to demonstrate how mobile technology can enhance financial management processes in higher education institutions, particularly within developing economies.

1.1. Subject and Field of Study

This research falls under the field of Information Technology and Software Engineering, with a focus on:

1. Mobile Application Development
2. Financial Technology (FinTech) Systems
3. Digital Payment Integration
4. Information Systems for Educational Institutions

The study combines research and system development to design and implement a mobile-based payment solution for higher education.

1.2. Study Objectives

1.2.1 Global (General) Objective

The general objective of this study is to design and development of financial mobile platform for students in Koforidua Technical University (KTU). And this could be achieved through the follow specific objectives.

1.2.2 Specific Objectives

The specific objectives of the project will be to:

1. Design and develop a module that will integrate multiple digital payment options.
2. Design and develop a module that will ensure real-time payment processing with instant confirmation notifications.
3. Design and develop a module that will provide transaction histories with downloadable PDF receipts.

1.3. Background to the Study

The process of fee payment in educational institutions has undergone significant transformation over time, particularly with the advancement of information and communication technologies (ICT). Traditionally, school fee payments were handled manually within individual institutions, where students or their parents were required to make payments directly at designated school offices or "tabletops." Upon payment, an invoice or receipt would be issued as proof of transaction. While this method was straightforward, it presented several challenges, including long queues, delays, and heavy administrative workload for school staff.

This manual approach often resulted in overcrowding within school premises, especially during peak periods such as the beginning of academic terms. Parents and guardians who accompanied their wards to school frequently spent long hours waiting to complete payments, sometimes returning home late due to the congestion. The inefficiencies associated with this system highlighted the need for a more structured and scalable payment method.

In response to these challenges, many institutions transitioned to a bank-based payment system. Under this approach, schools provided designated bank account details, requiring students or parents to visit specific banks to make fee payments. This shift reduced congestion within school environments and introduced a more formalized financial process. However, it also transferred the burden to the banking sector, where long queues became a common issue. Students occasionally had to request permission during academic hours to visit banks, while parents and guardians often spent considerable time waiting to complete transactions. Despite its improvements over the manual system, the bank-based method still suffered from inefficiencies and inconvenience.

With further advancements in digital financial services, some institutions began adopting mobile-based payment solutions, including Unstructured Supplementary Service Data (USSD) platforms. For example, Koforidua Technical University (KTU) has introduced USSD-based payment options to facilitate easier transactions. While this method improved accessibility and reduced dependence on physical locations, it introduced new challenges such as difficulty in remembering payment codes, dialing incorrect codes, and limited user interaction compared to modern applications.

These limitations indicate that, although progress has been made from manual to digital systems, existing solutions are still not fully optimized for user convenience, efficiency, and reliability. With the increasing penetration of smartphones among students, there is a strong opportunity to develop a more user-friendly and integrated mobile payment solution.

Therefore, this study proposes the development of a Student Pay Mobile Application that will enable students to conveniently make fee payments, access transaction history, and generate digital receipts through a simple and intuitive interface. This approach aims to address the shortcomings of previous systems while leveraging modern mobile technology to enhance efficiency, accuracy, and user experience in educational financial management.

1.4. Scope of Study

The scope of this study is limited to Koforidua Technical University. The project focuses on the design and development of a mobile application that will serve as an additional digital channel for school fee payments.

The system will:

1. Be accessible only to registered KTU students
2. Integrate mobile money and digital payment gateways
3. Provide transaction confirmation and digital receipts

The study does not include deployment to other institutions or integration with external university systems beyond payment processing.

1.5. Justification of the Study / Significance of the Study

This study is justified due to the increasing demand for digital financial solutions in higher education. Mobile payment systems enhance efficiency, reduce transaction costs, and improve transparency (Dahlberg et al., 2015).

Academically, the study contributes to research in mobile financial systems and information systems development within educational contexts. Practically, it offers a real-world solution to payment challenges faced by students and administrators at KTU.

The system is expected to:

1. Reduce congestion at banks during registration periods
2. Minimize risks associated with carrying cash
3. Improve financial record accuracy
4. Enhance transparency in university financial transactions

1.6. Methodology

This study adopts a combined research methodology and system development methodology to ensure both theoretical and practical objectives are achieved. The research component focuses on understanding the challenges of the current fee payment system, while the development component focuses on designing and implementing the proposed Student Pay Mobile Application.

1.6.1 Software Development Approaches

Although several software development approaches exist, including the Waterfall Model, Iterative and Incremental Models, Evolutionary Models, DevOps Approach, and Rapid

Application Development (RAD), the Agile methodology is considered the most suitable for the development of the proposed Student Pay Mobile Application.

The Waterfall Model follows a rigid and sequential process where each phase must be completed before moving to the next. This lack of flexibility makes it unsuitable for this project, as system requirements may evolve based on feedback from students and university staff during development. Similarly, while Iterative and Incremental Models allow partial system development, they do not emphasize continuous stakeholder collaboration to the same extent as Agile, which is critical for ensuring that the application meets real user needs.

Evolutionary Models such as the Spiral Model provide strong risk management but are often complex, costly, and time-consuming, making them less appropriate for an academic project with limited resources and time constraints. The DevOps Approach, although effective for continuous integration and deployment, is more suitable for large-scale production environments rather than a prototype or academic system under development. Likewise, Rapid Application Development (RAD) focuses on speed but may compromise system structure and long-term maintainability, especially when dealing with secure financial transactions.

In contrast, Agile methodology offers a flexible, iterative, and user-centered approach that is highly suitable for this project. Agile supports continuous feedback, allowing students and finance staff to actively participate in the development process and influence system improvements. It also enables incremental delivery of system components such as payment processing, receipt generation, and transaction history, ensuring that functional parts of the system are available early and can be tested frequently.

Furthermore, Agile promotes adaptability to changing requirements, reduces development risks through continuous testing, and improves overall system quality. Given the dynamic nature of user expectations and the need for a reliable and secure mobile payment system, Agile provides the most effective framework for achieving the objectives of this study within the given timeline.

1.6.2. Phases of Agile Methodology

The Agile methodology follows an iterative cycle rather than a strictly linear sequence. The key phases involved in Agile development include:

1. Requirement Gathering (Planning Phase)

In this phase, system requirements are identified through interactions with stakeholders such as students and finance staff. The requirements are documented as user stories and prioritized based on importance.

2. System Design

Based on the gathered requirements, the system architecture, database structure, and user interface designs are developed. This phase focuses on creating a blueprint for the system.

3. Development (Iteration/Sprint)

The system is developed in small units called sprints. Each sprint focuses on implementing specific features, such as payment integration or receipt generation.

4. Testing

Testing is conducted continuously during each sprint to ensure that the developed features function correctly. This includes unit testing, integration testing, and user acceptance testing.

5. Deployment

After successful testing, the developed features are deployed for use. In this project, deployment may involve making the application available for testing by selected users.

6. Review and Feedback

Stakeholders evaluate the deployed features and provide feedback. This feedback is used to improve the system in subsequent iterations.

7. Maintenance and Continuous Improvement

The system is continuously updated and improved based on user feedback and emerging requirements.

1.6.3. Tools and Technologies

The development of the Student Pay Mobile Application will utilize modern and widely adopted technologies to ensure efficiency, scalability, and maintainability.

1. Mobile Application Development

The mobile application will be developed using React Native.

Justification:

1. Enables cross-platform development (Android and iOS)
2. Reduces development time
3. Strong community support
4. High performance and reusable components

React Native allows a single codebase to serve multiple mobile platforms, making it cost-effective and efficient for academic projects.

2. Backend Development

The backend system will be developed using **PHP** in an API-based architecture.

Justification:

1. Lightweight and widely supported
2. Easy integration with databases
3. Suitable for RESTful API development
4. Strong compatibility with MySQL

The backend will handle authentication, transaction processing, receipt generation, and database interactions.

3. Database Management System

The database will be implemented using **MySQL**.

Justification:

1. Reliable relational database system
2. Efficient handling of structured data
3. Secure and scalable
4. Strong integration with PHP

MySQL will store student data, transaction records, payment logs, and receipt information.

4. Development Environment

The following code editors will be used:

1. **PhpStorm** – for backend development
2. **WebStorm** – for frontend/mobile development

These Integrated Development Environments (IDEs) provide advanced debugging tools, code intelligence, and improved productivity.

5. Testing and Deployment Tools

To expose the locally developed backend API to the internet for mobile application testing, **ngrok** will be used.

Justification:

1. Generates secure public URLs
2. Enables real-time mobile-to-backend testing
3. Simplifies development without full server deployment

1.7. Expected Results of the Study & Possible Use

The expected outcomes of the study include:

1. A functional Student Pay Mobile Application prototype
2. A secure backend system integrated with digital payment services
3. Real-time transaction confirmation system
4. Automated PDF receipt generation
5. Improved payment processing workflow

The system may serve as a model for other universities seeking to digitize their financial operations. It may also contribute to further research in educational FinTech solutions.

1.8. Presentation of Thesis (Chapter-by-Chapter Summary)

This thesis is organized into eight chapters, structured to reflect both the research and system development components of the study.

Chapter One: General Introduction (Research Proposal)

This chapter introduces the study by presenting the background of the research, problem statement, objectives (general and specific), scope of the study, justification, research methodology, software development approach, expected results, and work plan. It provides an overview of the entire research and system development process.

Chapter Two: Literature Review

This chapter reviews relevant literature on mobile payment systems, digital finance, financial technology (FinTech), and the integration of digital payment platforms in educational institutions. It also examines software development methodologies, particularly Agile methodology, and discusses theoretical frameworks related to technology adoption and system usability. The chapter identifies research gaps and establishes the foundation for the proposed system.

Chapter Three: Crystallization of the Research Problem

This chapter clearly defines and refines the research problem. It presents a detailed analysis of the existing fee payment system at Koforidua Technical University, identifies its weaknesses and limitations, and highlights the gap that the proposed system intends to fill. The chapter also presents research findings from interviews, questionnaires, observations, and desktop review.

Chapter Four: Analysis of the Proposed System

This chapter describes the functional and non-functional requirements of the proposed Student Pay Mobile Application. It includes feasibility analysis (technical, economic, and operational), system requirements specification, use case analysis, and an overview of the system architecture. The chapter explains how the proposed system will address the identified research problem.

Chapter Five: Detailed Design of the Proposed System

This chapter presents the technical design of the system. It includes database design (Entity-Relationship Diagram), system architecture diagrams, API design, payment processing workflow, user interface design, and security considerations. This chapter serves as the blueprint for system implementation.

Chapter Six: System Implementation and Testing

This chapter explains the development process of the system using the Agile Software Development Model. It describes the tools and technologies used, including React Native for the mobile application, PHP for backend API development, MySQL for database management, PhpStorm and WebStorm as development environments, and ngrok for backend testing. The chapter also presents system testing procedures and results, including unit testing, integration testing, and user acceptance testing.

Chapter Seven: System Documentation

This chapter provides technical and user documentation for the system. It includes installation procedures, system configuration, user guide (student interface), administrator guide, maintenance procedures, and backup strategies. This ensures the sustainability and usability of the developed system.

Chapter Eight: Conclusion and Recommendations

This chapter summarizes the study, highlights key findings, and evaluates whether the research objectives were achieved. It discusses the contributions of the project, outlines limitations encountered, and provides recommendations for future improvements and further research.

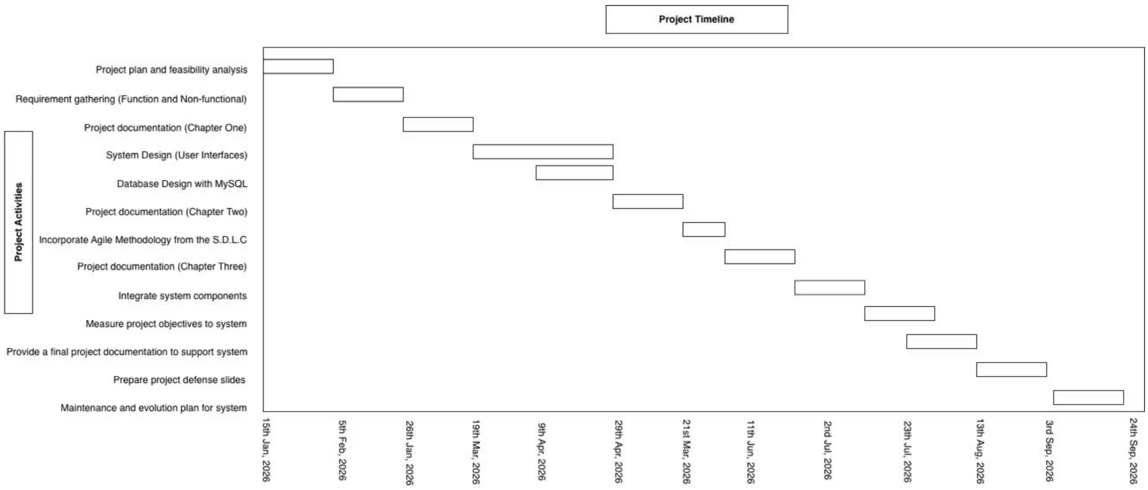
1.9. Study Work Plan (Timelines)

The project will follow a structured development schedule using a project management tool such as a **Gantt Chart** to plan and monitor activities. The major project activities will include:

1. Project plan and feasibility analysis
2. Requirement gathering (Function and Non-functional)
3. Project documentation (Chapter One)
4. System Design (User Interfaces)
5. Database Design with MySQL
6. Project documentation (Chapter Two)
7. Incorporate Agile Methodology from the S.D.L.C
8. Project documentation (Chapter Three)
9. Integrate system components
10. Measure project objectives to system
11. Provide a final project documentation to support system

12. Prepare project defence slides
13. Maintenance and evolution plan for system

The Gantt Chart will be used to allocate timelines, track progress, and ensure the project is completed within the scheduled timeframe.



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